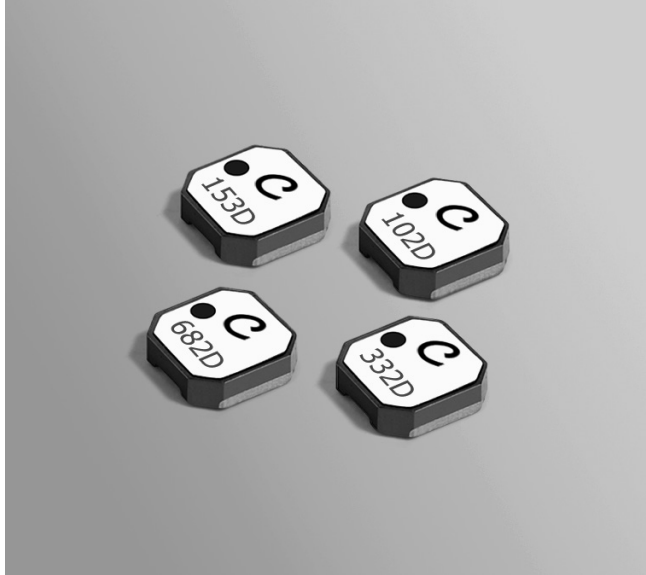


Shielded Power Inductors – LPS4012



- Very low DCR; excellent current handling
- 4.0 × 4.0 mm footprint; less than 1.2 mm tall
- AEC-Q200 Grade 1 qualified (–40°C to +125°C ambient)

Core material Ferrite

Core and winding loss See www.coilcraft.com/coreloss

Environmental RoHS compliant, halogen free

Terminations RoHS compliant matte tin over nickel over silver. Other terminations available at additional cost.

Weight 54 – 64 mg

Ambient temperature –40°C to +85°C with (40°C rise) Irms current.

Maximum part temperature +125°C (ambient + temp rise). [Derating](#).

Storage temperature Component: –40°C to +125°C.

Tape and reel packaging: –40°C to +80°C

Resistance to soldering heat Max three 40 second reflows at +260°C, parts cooled to room temperature between cycles

Moisture Sensitivity Level (MSL) 1 (unlimited floor life at <30°C / 85% relative humidity)

Failures in Time (FIT) / Mean Time Between Failures (MTBF)

38 per billion hours / 26,315,789 hours, calculated per Telcordia SR-332

Recommended pick and place nozzle OD: 4 mm; ID: ≤ 2 mm

PCB washing Tested to MIL-STD-202 Method 215 plus an additional aqueous wash. See [Doc787_PCB_Washing.pdf](#).

Part number ¹	Inductance ² (µH)	DCR max ³ (Ohms)	SRF typ ⁴ (MHz)	Isat (A) ⁵			Irms (A) ⁶	
				10% drop	20% drop	30% drop	20°C rise	40°C rise
LPS4012-331MR_	0.33±20%	0.025	375	5.2	5.4	5.5	2.2	3.0
LPS4012-681MR_	0.68±20%	0.055	220	3.5	3.6	3.7	1.8	2.4
LPS4012-102NR_	1.0±30%	0.060	180	2.8	2.9	3.0	1.7	2.4
LPS4012-152MR_	1.5±20%	0.070	140	2.6	2.7	2.8	1.6	2.2
LPS4012-222MR_	2.2±20%	0.100	115	2.3	2.4	2.5	1.2	1.75
LPS4012-332MR_	3.3±20%	0.100	100	1.3	1.4	1.4	1.45	2.00
LPS4012-472MR_	4.7±20%	0.175	70	1.6	1.7	1.8	1.10	1.45
LPS4012-562MR_	5.6±20%	0.260	60	1.5	1.6	1.6	0.85	1.10
LPS4012-682MR_	6.8±20%	0.340	55	1.3	1.3	1.4	0.80	0.98
LPS4012-103MR_	10±20%	0.350	40	0.98	1.0	1.1	0.55	0.75
LPS4012-153MR_	15±20%	0.550	30	0.79	0.82	0.84	0.53	0.73
LPS4012-223MR_	22±20%	0.600	25	0.74	0.78	0.79	0.52	0.70
LPS4012-333MR_	33±20%	0.825	22	0.45	0.47	0.48	0.46	0.61
LPS4012-473MR_	47±20%	1.40	19	0.35	0.37	0.38	0.40	0.52
LPS4012-683MR_	68±20%	1.70	15	0.30	0.32	0.33	0.35	0.46
LPS4012-104MR_	100±20%	2.40	12	0.24	0.26	0.27	0.30	0.40
LPS4012-124MR_	120±20%	3.30	11.5	0.23	0.24	0.25	0.27	0.36
LPS4012-154MR_	150±20%	3.50	10.0	0.21	0.22	0.23	0.25	0.32
LPS4012-184MR_	180±20%	5.00	8.0	0.18	0.19	0.20	0.23	0.29
LPS4012-224MR_	220±20%	5.20	7.0	0.15	0.16	0.17	0.21	0.27
LPS4012-334MR_	330±20%	7.20	7.0	0.14	0.14	0.15	0.17	0.225
LPS4012-474MR_	470±20%	10.0	4.0	0.10	0.11	0.12	0.13	0.175
LPS4012-564MR_	560±20%	12.5	3.5	0.10	0.105	0.115	0.11	0.140
LPS4012-684MR_	680±20%	13.5	3.0	0.10	0.105	0.110	0.11	0.135
LPS4012-824MR_	820±20%	20.0	3.0	0.090	0.095	0.095	0.105	0.132
LPS4012-105MR_	1000±20%	21.5	3.0	0.080	0.090	0.095	0.100	0.130
LPS4012-155MR_	1500±20%	30.0	2.5	0.080	0.090	0.090	0.087	0.115
LPS4012-185MR_	1800±20%	35.0	2.0	0.079	0.085	0.087	0.075	0.100
LPS4012-225MR_	2200±20%	42.0	1.0	0.079	0.083	0.085	0.070	0.090

1. Please specify **termination** and **packaging** codes:

LPS4012-682MRC

Termination: R= RoHS compliant matte tin over nickel over silver.

Special order, added cost:

Q = RoHS tin-silver-copper (95.5/4/0.5)

or **P** = non-RoHS tin-lead (63/37).

Packaging: C= 7" machine-ready reel. EIA-481 embossed plastic tape (1000 parts per full reel).

B= Less than full reel. In tape, but not machine ready.

To have a leader and trailer added (\$25 charge), use code letter C instead.

D= 13" machine-ready reel. EIA-481 embossed plastic tape. Factory order only, not stocked (3500 parts per full reel).

2. Inductance tested at 100 kHz, 0.1 Vrms using an Agilent/HP 4192A. Inductance at 1 MHz is the same for parts with SRF ≥10 MHz.

3. DCR measured on a micro-ohmmeter.

4. SRF measured using Agilent/HP 8753ES or equivalent.

5. DC current at 25°C that causes the specified inductance drop from its value without current.

[Click for temperature derating information.](#)

6. Current that causes the specified temperature rise from 25°C ambient. This information is for reference only and does not represent absolute maximum ratings.

[Click for temperature derating information.](#)

7. Electrical specifications at 25°C.

Refer to Doc 362 "Soldering Surface Mount Components" before soldering.



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Document 433-1 Revised 10/01/19

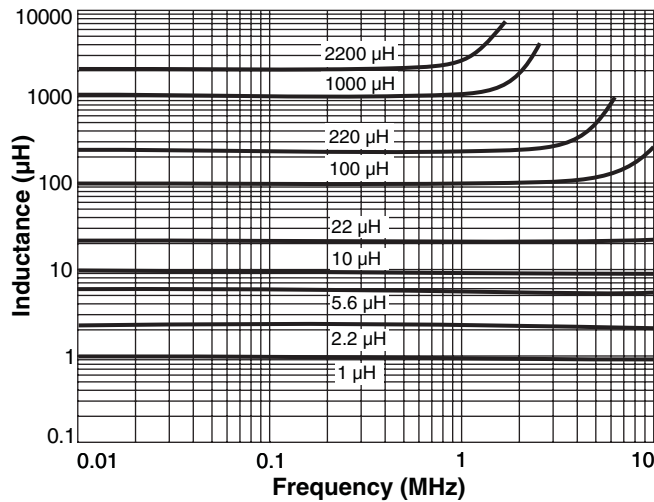
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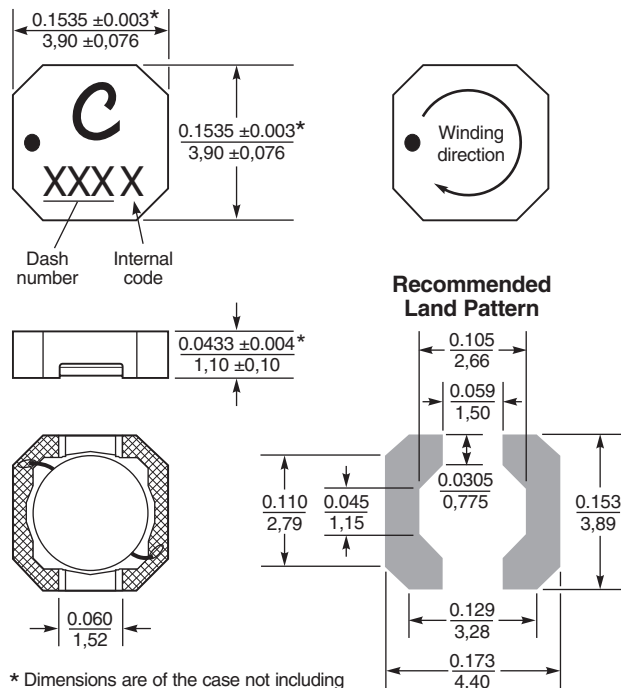
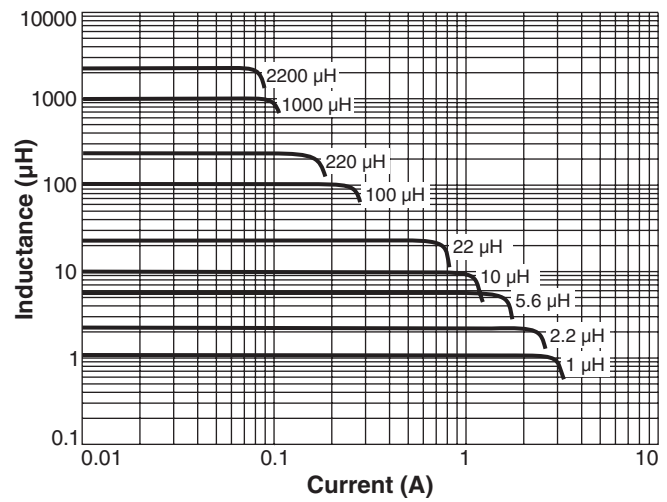


Shielded SMT Power Inductors – LPS4012 Series

Typical L vs Frequency



Typical L vs Current

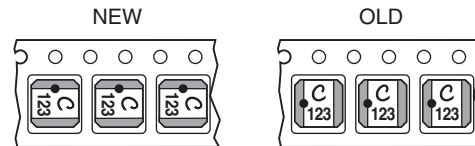


* Dimensions are of the case not including the termination. For maximum overall dimensions including the termination, add 0.005 in / 0.13 mm.
For optional tin-lead and tin-silver-copper terminations, dimensions are for the mounted part. Dimensions before mounting can be an additional 0.005 inch / 0.13 mm).

Dimensions are in $\frac{\text{inches}}{\text{mm}}$

Packaging 1000/7" reel; 3500/13" reel Plastic tape: 12 mm wide, 0.25 mm thick, 8 mm pocket spacing, 1.32 mm pocket depth

NOTE NEW PART ORIENTATION Parts are rotated 90° in the packaging tape compared to previous versions of this product.



Coilcraft **Designer's Kit C392** contains samples of 0.56 μH to 33 μH parts (3 each) from LPS3008, LPS3010 and LPS3015. **Kit C401** contains samples of 0.56 μH to 33 μH parts (3 each) from LPS4012 and LPS4018. **Kit C402** contains samples of 220 μH to 2200 μH parts (3 each) from all five series. For details of kit contents and to order, contact Coilcraft or visit <http://order.coilcraft.com>.

Packaging 1000/7" reel; 3500/13" reel Plastic tape: 12 mm wide, 0.25 mm thick, 8 mm pocket spacing, 1.32 mm pocket depth



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Document 433-2 Revised 10/01/19

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Mouser Electronics

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